

SIBIU, Romania

WMO: 152600

Lat: 45.7894N

Lon: 24.0914E

Elev: 1458

StdP: 13.94

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	8.9	-0.5	5.6	4.5	5.3	7.6	10.7	22.9	41.5	19.9	41.9	2.0	220	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	88.3	67.5	85.3	67.0	82.3	65.8	70.6	82.4	69.0	80.5	67.5	78.6	8.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	104.2	74.9	65.1	98.1	73.4	63.8	93.4	72.1	35.4	82.6	33.9	80.6	32.7	78.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.3	17.1	14.2	DB	-6.3	93.0	7.2	3.9	-11.5	95.8	-15.7	98.1	-19.8	100.2	-25.0	103.1	(4)
			WB	-6.6	73.5	7.2	1.9	-11.8	74.9	-16.0	76.0	-20.0	77.1	-25.3	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	28.1	32.9	40.7	50.9	59.4	65.6	68.6	68.4	59.8	50.1	40.7	30.6	(6)
		DBStd	16.11	9.34	9.56	8.21	7.08	5.95	5.69	5.08	5.16	6.51	7.39	8.62	9.71	(7)
		HDD50	2533	679	480	299	72	6	0	0	4	92	298	603		(8)
		HDD65	5959	1144	899	753	425	189	61	24	25	180	464	728	1067	(9)
		CDD50	2431	0	1	11	98	295	469	577	569	297	94	20	0	(10)
		CDD65	383	0	0	0	1	14	79	136	129	23	1	0	0	(11)
		CDH74	4012	0	0	1	30	214	816	1332	1300	296	23	0	0	(12)
		CDH80	1186	0	0	0	2	24	213	432	445	67	3	0	0	(13)
(14)	Wind	WSAvg	4.7	3.7	4.4	5.6	5.7	5.2	4.9	4.9	4.5	4.6	4.1	4.4	3.8	(14)
(15)	Precipitation	PrecAvg	25.7	1.1	1.0	1.3	2.1	3.1	3.9	3.7	2.8	2.1	1.7	1.3	1.2	(15)
(16)		PrecMax	34.4	2.8	2.1	3.4	5.1	6.7	9.2	8.9	8.9	7.3	5.8	3.6	3.2	(16)
(17)		PrecMin	14.2	0.2	0.0	0.1	0.2	1.2	0.1	0.5	0.1	0.0	0.1	0.0	0.0	(17)
(18)		PrecStd	4.4	0.6	0.6	0.8	1.1	1.5	1.9	1.7	1.9	1.5	1.1	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	53.8	60.5	68.3	77.9	82.8	89.1	93.1	92.9	86.5	76.6	68.3	57.0		(19)
(20)		0.4% MCWB	46.0	48.7	51.1	58.6	62.5	67.9	67.3	67.3	63.8	60.2	54.7	47.9		(20)
(21)		2% DB	48.8	54.5	63.4	72.1	79.1	85.6	88.1	88.7	81.1	72.5	63.5	51.8		(21)
(22)		2% MCWB	43.6	45.6	49.3	55.9	61.8	67.6	68.0	67.7	62.9	58.4	52.8	45.3		(22)
(23)		5% DB	44.3	50.4	59.0	68.0	76.2	82.2	84.7	84.9	77.1	68.1	59.4	47.2		(23)
(24)		5% MCWB	40.3	43.3	47.4	53.8	61.1	66.2	67.6	66.9	61.3	57.0	50.9	42.7		(24)
(25)		10% DB	40.2	46.4	54.9	64.5	72.8	78.9	81.4	81.3	73.2	63.7	55.1	43.1		(25)
(26)		10% MCWB	37.3	41.1	45.6	52.0	59.5	65.0	66.3	65.5	60.5	54.7	48.6	40.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	46.8	49.4	53.2	60.2	66.6	72.2	72.5	72.9	67.2	63.1	56.3	48.9		(27)
(28)		0.4% MCDB	52.3	58.5	65.3	74.5	78.0	83.6	84.2	85.1	80.0	73.8	65.0	54.8		(28)
(29)		2% WB	43.7	46.4	50.4	57.6	64.0	69.4	70.5	70.4	64.5	60.0	53.7	45.8		(29)
(30)		2% MCDB	48.4	53.6	60.7	69.7	74.9	80.9	81.9	82.8	76.4	69.2	61.9	51.2		(30)
(31)		5% WB	40.5	44.1	48.3	55.3	62.1	67.6	69.1	68.4	62.9	57.8	51.5	42.9		(31)
(32)		5% MCDB	43.8	49.5	57.9	65.7	73.1	78.9	80.4	79.7	74.0	66.1	58.7	47.0		(32)
(33)		10% WB	37.5	41.5	46.1	53.2	60.5	66.0	67.6	66.9	61.3	55.6	49.0	40.5		(33)
(34)		10% MCDB	39.7	46.1	53.9	61.9	70.9	76.8	78.5	77.9	70.8	62.8	54.7	43.0		(34)
(35)	Mean Daily Temperature Range	MDBR	14.2	16.6	18.9	20.5	20.8	21.0	21.6	22.4	20.8	21.3	17.9	13.4		(35)
(36)		5% DB MCDBR	18.2	22.7	27.2	26.9	26.8	26.6	27.2	28.2	27.8	28.6	24.9	19.0		(36)
(37)		5% DB MCWBR	13.2	14.5	15.0	13.4	12.9	11.9	11.3	11.0	12.1	15.9	15.2	13.2		(37)
(38)		5% WB MCDBR	17.2	21.0	24.7	24.2	24.5	24.5	23.9	24.7	24.6	25.4	23.0	17.6		(38)
(39)		5% WB MCWBR	12.8	13.8	14.2	12.5	12.8	11.9	11.3	11.3	11.8	14.9	14.4	12.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.337	0.379	0.439	0.424	0.420	0.439	0.450	0.403	0.364	0.337	0.308		(40)
(41)		taud	2.345	2.276	2.225	2.096	2.210	2.257	2.212	2.174	2.275	2.380	2.399	2.376		(41)
(42)		Ebn at Noon	250	263	267	259	268	269	262	253	256	251	237	236		(42)
(43)		Edh at Noon	24	31	38	48	44	42	43	44	36	28	23	21		(43)
(44)	All-Sky Solar Radiation	RadAvg	423	682	1028	1366	1653	1814	1812	1652	1191	827	482	341		(44)
(45)		RadStd	49	84	125	148	158	166	156	151	124	101	56	42		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.42	N/A	N/A	N/A	N/A	N/A	-272	-398	+216	+90	(46)
(47)	Regional (75 neighbors)	+1.21	N/A	N/A	+1.60	N/A	N/A	-241	-351	+153	+34	(47)

Nomenclature: See separate page